WAZUH SOC LAB REPORT

SIEM and Endpoint Monitoring Fundamentals

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Course: SIEM and Endpoint Monitoring Fundamentals

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1. Introduction

Security Information and Event Management (SIEM) systems help security teams collect, monitor, and analyse security events from different systems. They provide visibility into activities that may indicate security incidents and help analysts investigate suspicious behaviour.

In this lab, we focused on Wazuh file integrity monitoring on a Windows endpoint. File Integrity Monitoring (FIM) helps detect changes to files and directories by recording activities such as file creation, modification, and deletion.

The exercise involved verifying the Wazuh services, accessing the dashboard, connecting a Windows endpoint, configuring real-time file monitoring, and testing whether Wazuh could detect changes to the file.

2. Objectives

The objectives of this lab were to:

1. Verify that the Wazuh Manager, Indexer, and Dashboard services were running.
2. Access the Wazuh Dashboard and confirm endpoint visibility.
3. Register and verify a Windows endpoint agent.
4. Configure real-time File Integrity Monitoring on a sensitive directory.
5. Test file creation, modification, and deletion.
6. Analyse the generated events, file metadata, checksums, and alert levels.
7. Understand how FIM supports security monitoring and incident investigation.

3. Lab Environment and Architecture

The lab used an Ubuntu virtual machine provided by the academy and a Windows 11 endpoint. Both systems were configured in a VirtualBox host-only network.

The Ubuntu server hosted the Wazuh components, while the Windows endpoint ran the Wazuh agent. I was able to access the Wazuh dashboard through a web browser.

Component	Details
Wazuh server	`icdfa-server`
Server operating system	Ubuntu
Wazuh Manager	Pre-deployed by the academy
Wazuh Indexer	Pre-deployed by the academy
Wazuh Dashboard	Pre-deployed by the academy
Endpoint	Windows 11 Pro Education N
Endpoint IP address	`192.168.56.112`
Agent name	`JayAfo_PC`
Agent ID	`005`
Wazuh agent version	`4.12.0`
Virtualisation platform	VirtualBox
Network configuration	Host-only
Monitored directory	`C:\Users\Juliet\Documents\SensitiveFiles`
Test file	`secret_plan.txt`

Note: The Wazuh environment was provided as a pre-deployed academy server. The lab involved verifying and using the existing deployment rather than installing all Wazuh components from scratch.

4. Phase 1: Wazuh Service Verification

4.1 Procedure

The Ubuntu server was started, and the status of the Wazuh services was checked using the Linux terminal.

The following commands were used:

```
sudo systemctl status wazuh-manager
sudo systemctl status wazuh-indexer
sudo systemctl status wazuh-dashboard
```

4.2 Results

The commands returned the following results:

Service	Result
Wazuh Manager	Active
Wazuh Indexer	Active
Wazuh Dashboard	Active

All three services were active, indicating that the main Wazuh components were running and ready for use.

Conclusion: The service verification was successful.

Evidence 1: Wazuh_Dashboard_Status, Wazuh_Indexer_Status, Wazuh_Status

5. Phase 2: Dashboard Access and Verification

5.1 Procedure

The Wazuh Dashboard was opened in a web browser on the Windows 11 endpoint. This was done by navigating to the Wazuh server IP address **(192.168.56.110).** The dashboard was checked to confirm that it was accessible and that the registered Windows agent was visible.

The agent status was reviewed before proceeding with the file integrity tests.

5.2 Results

The dashboard loaded successfully, and the Windows endpoint appeared in the agent list with an Active status.

No relevant alerts were visible during the initial dashboard check. The subsequent FIM tests generated events that were reviewed during the investigation.

Conclusion: Dashboard access was successful, and the connected endpoint was visible.

Evidence 2: Wazuh_Agent_Active

6. Phase 3: Windows Agent Registration and Verification

6.1 Procedure

The Windows endpoint was connected to the Wazuh server through the configured agent. The dashboard was used to verify the agent's registration, connection status, and system details.

6.2 Agent Details

The following information was displayed in the Wazuh Dashboard:

Field	Value
Agent name	`JayAfo_PC`
Agent ID	`005`

Status	Active
IP address	`192.168.56.112`
Operating system	Microsoft Windows 11 Pro Education N
OS version	`10.0.26300.9711`
Agent version	`4.12.0`
Agent group	`default`
Cluster node	`node01`
Registration date	17 October 2026, 18:49:18

The agent was active and communicating with the Wazuh environment. This confirmed that the endpoint was available for monitoring.

Conclusion: Windows agent verification was successful.

Evidence 3: Registering_Wazuh_Agent, Wazuh_Agent_Active

7. Phase 4: File Integrity Monitoring Configuration

7.1 Objective

The purpose of this phase was to configure Wazuh to monitor a sensitive directory on the Windows endpoint and detect changes to files stored inside it.

7.2 Procedure

A directory named `SensitiveFiles` was used for the exercise.

The full path was:

```
C:\Users\Juliet\Documents\SensitiveFiles
```

The Wazuh agent configuration file, `ossec.conf`, was updated with a monitoring entry for the directory. Real-time monitoring was enabled using the `realtime="yes"` setting.

The agent was then restarted through the Wazuh Agent Manager interface to apply the configuration.

After the restart, the agent was checked in the dashboard and remained Active.

7.3 Result

The configuration was applied successfully. Subsequent tests confirmed that Wazuh detected file creation, modification, and deletion within the monitored directory.

Conclusion: The FIM configuration was operational.

Evidence 4: ossec.conf_edit

8. Phase 5: File Integrity Monitoring Tests

Three tests were performed to verify whether Wazuh could detect changes to a file in the monitored directory.

The test file was:

`C:\Users\Juliet\Documents\SensitiveFiles\secret_plan.txt`

8.1 File Creation Test

Procedure

A text file named **`secret_plan.txt`** was created in the monitored directory. The Wazuh Dashboard was then checked for a corresponding file creation event.

Results

Wazuh recorded the event with the following details:

Field	Result
Event description	File added to the system
Rule ID	**`554`**
Alert level	**`5`**

File	`secret_plan.txt`
Monitoring mode	Real-time

The event confirmed that Wazuh detected the addition of the new file to the monitored directory.

Conclusion: The file creation test was successful.

Evidence 5: Registering_Wazuh_Agent

Evidence 6: EmptyFile_Created_Details

8.2 File Modification Test

Procedure

After creating the file, additional text was appended to the existing content. The file was saved, and the Wazuh Dashboard was refreshed to check for modification events.

Results

Wazuh detected the change and generated an event with the decoder `syscheck_integrity_changed`.

The event contained the following information:

Field	Original value	New value
File size	93 bytes	163 bytes
MD5	`a160ca83a68b64f95e6cb7f5d8adca41`	`4bbb1098ddf87299ee822392d272d937`
SHA1	`399306ff888f56fba3b535fef524514fb05ced62`	`9fbfc7c655497be7940075464fb6ed0507ddff52`
SHA256	`9df8c07bcfa0985133648aea62f537f6ec590bcf7f369ba23a8dbc12c3639d27`	`44c2923604133fe006b99f9eeee3adf3efe6fc854faa43a8157b40a429adcb59`

Additional event details included:

- **Event description:** Integrity checksum changed.

- **Rule ID:** `550`
- **Alert level:** `7`
- **Monitoring mode:** Real-time.
- **Manager:** `icdfa-server`
- **Agent:** `JayAfo_PC`

The file size increased from 93 bytes to 163 bytes, and all three checksum values changed. This confirmed that the file's contents had been modified.

Analysis

Checksums provide a way to identify changes to file contents. When the contents of a file change, its checksum will generally change as well. Wazuh recorded the old and new checksum values, allowing the analyst to compare the file's state before and after modification.

Conclusion: The file modification test was successful. Wazuh detected the changes and recorded the updated file metadata and checksums.

Evidence 7: FIM_Text, First_Text_Modify_Test, Second_Text_Modify_Test.

8.3 File Deletion Test

Procedure

After confirming the modification event, the test file was deleted from the monitored directory. The Wazuh Dashboard was then checked for a file deletion event.

Results

Wazuh detected the deletion in real time.

The event details were:

Field	Result
Decoder	`syscheck_deleted`
Event description	File deleted
Rule ID	`553`
Alert level	`7`

Monitoring mode	Real-time
Agent name	`JayAfo_PC`
Agent ID	`005`
Manager	`icdfa-server`
File path	`c:\users\juliet\documents\sensitivefiles\secret_plan.txt`
MITRE ATT&CK technique	`T1070.004`, `T1485`

The event confirmed that Wazuh detected and recorded the deletion of the monitored file.

The associated MITRE ATT&CK mappings included File Deletion and Data Destruction. These mappings provide security context for the event, although a file deletion alone does not prove malicious activity.

Conclusion: The file deletion test was successful.

Evidence 8: file_deletion_proof1, file_deletion_proof2.

8.4 Summary of FIM Test Results

Test	Event description	Rule ID	Alert level	Result
File creation	File added to the system	554	5	Successful
File modification	Integrity checksum changed	550	7	Successful
File deletion	File deleted	553	7	Successful

All three tests were successful. Wazuh detected the file creation, modification, and deletion activities in the monitored directory.

The results demonstrated that the configured real-time File Integrity Monitoring was working as expected.

9. Phase 6: Analysis and Discussion

9.1 What Metadata and Checksum Information Did Wazuh Record?

Wazuh recorded information about the monitored file, including its path, size, modification time, and checksum values. It also generated events when the file was created, modified, and deleted.

During the modification test, the file size increased from 93 bytes to 163 bytes. Its MD5, SHA1, and SHA256 checksums also changed.

This information helps a security analyst identify changes to files and investigate potentially suspicious activity. The recorded events provide evidence of what happened to the monitored file during the test.

9.2 Which Two Directories Might Generate Excessive FIM Alerts?

Two examples of directories that may generate excessive FIM alerts are:

1. Windows temporary directory (`C:\Windows\Temp`)

Windows and installed applications use temporary files while performing routine operations, installing updates, and running programs. These files may be created, modified, and deleted frequently.

Monitoring this directory may generate many alerts for normal activity, making it more difficult to identify important security events.

2. Browser cache directories

Web browsers store cached images, scripts, and other website resources in user-profile directories. These files change regularly as users browse websites.

Monitoring browser cache directories may therefore generate many alerts even when no malicious activity is taking place.

These are examples of potentially noisy directories, not directories tested during this lab. In a real environment, monitoring should be based on the organisation's security requirements. Exclusions should be considered carefully because excluding a directory may also reduce visibility into malicious activity.

9.3 How Can FIM Help Detect Ransomware Activity?

Ransomware may encrypt, modify, rename, or delete files. FIM can detect changes to monitored files and record information such as file paths, modification times, and checksum values.

If a large number of files are modified or deleted within a short period, the activity may indicate a ransomware attack and require further investigation.

However, file changes alone do not confirm ransomware activity. An analyst would need to investigate the surrounding events and use additional security tools, such as endpoint protection and other monitoring capabilities, to determine what happened.

In this lab, normal file operations were tested. Ransomware was not executed.

9.4 How Does Wazuh FIM Compare with Native Windows Event Logs?

Wazuh FIM provides dedicated monitoring for configured files and directories. It can record file changes, compare checksum values, assign alert levels, and present events through a centralized dashboard.

Windows Event Logs provide useful operating-system, application, and security records. However, file-change auditing and centralised collection may require additional Windows configuration or other tools.

Wazuh and Windows Event Logs can complement each other. Windows Event Logs provide information about system activities, while Wazuh helps security analysts monitor file integrity, organise security events, and investigate changes across connected endpoints.

10. Challenges Encountered and Resolutions

During the exercise, I initially had difficulty locating the relevant file integrity events in the Wazuh Dashboard. Although the events were being generated, I needed to navigate to the appropriate event view and search for the correct file path.

Another task was configuring the monitored directory and restarting the Wazuh agent before performing the tests.

These issues were resolved by checking the agent status, reviewing the event details, and searching for the monitored file path.

After these steps, I successfully located the file creation, modification, and deletion events.

11. Conclusion

This lab provided practical experience with Wazuh and File Integrity Monitoring. I verified the status of the Wazuh services, accessed the dashboard, confirmed the Windows agent connection, and configured real-time monitoring for a test directory.

The tests confirmed that Wazuh detected file creation, modification, and deletion. The modification event also provided evidence of changes to the file's size and checksum values.

The exercise improved my understanding of how FIM can support endpoint monitoring, security investigations, and the detection of suspicious file activity.

12. Recommendations

Based on the exercise, the following recommendations are proposed:

1. Monitor sensitive files and directories that are important to the organisation.
2. Review FIM alerts regularly and investigate unexpected file changes.
3. Configure monitoring carefully to reduce unnecessary alerts without creating security gaps.
4. Combine Wazuh with Windows security logs and endpoint protection tools for better visibility.
5. Keep Wazuh agents and server components properly configured and updated.
6. Retain relevant event records and screenshots to support security investigations and audits.
7. Investigate unusual patterns of file modification or deletion rather than treating every file change as a confirmed security incident.